# SORTING WORKCELL

DEMO COMMAND SHEET | ROS 2 JAZZY + GAZEBO HARMONIC

## 1 Start the bridge - Terminal 1 (WSL Ubuntu)

```
source /opt/ros/jazzy/setup.bash
ros2 launch rosbridge_server rosbridge_websocket_launch.xml
```

Leave this terminal open. It serves the dashboard connection at `ws://localhost:9090`.

## 2 Serve the dashboard - Terminal 2 (WSL Ubuntu)

```
cd ~/robotics_ws
python3 -m http.server 8000 --directory web --bind 0.0.0.0
```

Leave this terminal open. In Windows Chrome, open `http://localhost:8000`.

## 3 Choose a cube colour and start the stack - Terminal 3 (WSL Ubuntu)

```
source /opt/ros/jazzy/setup.bash
bash ~/robotics_ws/src/robot_description/worlds/switch_demo_cube.sh red
bash ~/robotics_ws/restart_sorting_workcell.sh
```

Use `red`, `blue`, or `green` on the second line. Restart rebuilds both ROS packages and starts Gazebo, camera/clock bridges, perception, and task manager. Keep rosbridge and the web server running.

## 4 Run the live demo - Browser

Refresh `http://localhost:8000`. Click **Connect**; wait for Connected to rosbridge, a live camera frame, and joint positions. Click **Start** once. Let the cycle finish; confirm the object is in the matching colour bin and the arm returns home.

To demo another colour, repeat Step 3 with `blue` or `green`, then refresh the page and start one cycle.

## Quick checks if the page says no camera / no detection - Terminal 4

```
source /opt/ros/jazzy/setup.bash
source ~/robotics_ws/install/setup.bash
ros2 topic list | grep -E '/camera/image_raw|/joint_states|/detected_object|/sorting/status'
ros2 topic hz /camera/image_raw
```

Camera topic should appear and report a non-zero rate. Press `Ctrl+C` to stop the rate check. In the browser, make sure the dashed pickup region contains the cube.